① Linear Equation : $ax_1 + bx_2 + cx_3 = D$

(a: coefficient; D: constant number; x_1: variable : power of 1)

~~x^{-2}~~

② Linear System or System of Linear equations :

$$x_1 - 3x_2 + 0x_3 = 5$$
$$-x_1 + x_2 + 5x_3 = 2$$
$$0x_1 + x_2 + x_3 = 0$$

③ Matrix : $m \times n$ matrix (m: no. of rows; n: no. of columns)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

entry (entries)

2×3 matrix

Augmented matrix :

$$\begin{array}{c} \begin{matrix} x_1 & x_2 & x_3 & C \end{matrix} \\ \left[\begin{array}{cccc} 1 & -3 & 0 & 5 \\ -1 & 1 & 5 & 2 \\ 0 & 1 & 1 & 0 \end{array}\right] \end{array} \begin{matrix} \text{L.E1 / R1} \\ \text{L.E2 / R2} \\ \text{L.E3 / R3} \end{matrix}$$

Coefficient matrix :

$$\begin{bmatrix} 1 & -3 & 0 \\ -1 & 1 & 5 \\ 0 & 1 & 1 \end{bmatrix}$$

④ Row reduce the linear system (in ②) by using Elementary Row Operation.

- Replacement : $R_1 = R_1 + 2R_2$
- Scaling : $R_2 = \frac{1}{5}R_2$
- Interchange : $R_1 \leftrightarrow R_2$

augmented matrix.

$$\begin{array}{c} \begin{matrix} x_1 & x_2 & x_3 & C \end{matrix} \\ \left[\begin{array}{cccc} 1 & -3 & 0 & 5 \\ -1 & 2 & 5 & 2 \\ 0 & 1 & 3 & 0 \end{array}\right] \end{array} \begin{matrix} \text{L.E1 / R1} \\ \text{L.E2 / R2} \\ \text{L.E3 / R3} \end{matrix}$$

$\downarrow R_2 = R_2 + R_1$

$$\left[\begin{array}{cccc} 1 & -3 & 0 & 5 \\ 0 & -1 & 5 & 7 \\ 0 & 1 & 3 & 0 \end{array}\right]$$

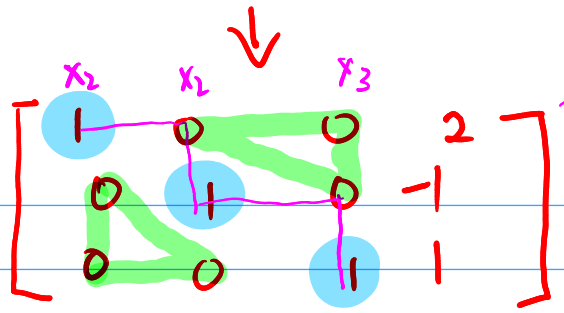
$\rightarrow$ $\left[\begin{array}{ccc|c} 1 & 0 & 0 & ? \\ 0 & 1 & 0 & ? \\ 0 & 0 & 1 & ? \end{array}\right]$

$\left[\begin{array}{ccc|c} 2 & 0 & 1 & 10 \\ 0 & 1 & 2 & 10 \\ 0 & 0 & & 10 \end{array}\right]$ or $\left[\begin{array}{ccc|c} 10 & 9 & 9 & \\ 0 & 1 & 10 & \\ 0 & 0 & 00 & \end{array}\right]$

$\downarrow$

⋮

Echelon form:
→ nonzero leading entries.
→ the no. below the leading entries must be '0'



Reduced echelon form.
→ leading entry of '1' is a pivot.
→ the numbers above & below the pivot : 0
→ the pivots form: ⌞

→ ⑤ Find the solutions for the linear system (2).

$x_1 = 2$
$x_2 = -1$
$x_3 = 1$

or $(x_1, x_2, x_3) = (2, -1, 1)$ *.

⑥

Reduced echelon matrix	solutions.	System consistent/ inconsistent.
$\left[\begin{array}{cccc} x_1 & x_2 & x_3 & c \\ 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 5 \end{array}\right]$ (augmented matrix)	$x_1 = 10$ $x_2 = 2$ $x_3 = 5$ Exactly one solution	consistent.
$\left[\begin{array}{cccc} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 5 \end{array}\right]$ $0 = 5$	$x_1 = 10$ $x_2 = 2$ $x_3 = X$ NO solution	inconsistent.
$\left[\begin{array}{cccc} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$ (pivot, pivot; x_2 x_1 x_3)	infinitely many solutions	consistent.

Basic variable
$x_1 = 10$
$x_2 = 2$
$0 = 0$

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \\ 0 + x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\vec{x}$: vector in $\mathbb{R}^3$

x_3: Free variable (any numbers)

when $x_3 = 0$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix}$

$x_3 = 1$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \\ 1 \end{bmatrix}$

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